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Geometric Transformations Using OpenCV

Aim

To write a Python program using OpenCV to perform various geometric transformations on an image.

The program performs the following operations:

- Image Translation
- Image Scaling (Resizing)
- Image Shearing
- Image Reflection (Flipping)
- Image Rotation

Software Used

- Anaconda – Python 3.7
- Jupyter Notebook / VS Code
- OpenCV (`cv2`)
- NumPy
- Matplotlib

Algorithm

Step 1:

Import the required libraries: OpenCV, NumPy, and Matplotlib.

Step 2:

Read the input image in color mode.

Step 3: Image Translation

- Create a translation matrix to shift the image
- Move the image 50 pixels to the right and 80 pixels down
- Apply transformation using `cv2.warpAffine()`
- Display original and translated images

Step 4: Image Scaling

- Resize the image to 0.5× (downscale)
- Resize the image to 2× (upscale)
- Use `cv2.resize()`
- Display original, downscaled, and upscaled images

Step 5: Image Shearing

- Create transformation matrices for:
  - Horizontal shearing
  - Vertical shearing
- Apply transformations using `cv2.warpAffine()`
- Display original and sheared images

Step 6: Image Reflection

- Perform flipping using `cv2.flip()` :
  - Horizontal reflection
  - Vertical reflection

- Both axes
- Display all reflected images

Step 7: Image Rotation

- Create rotation matrices for:
  - 45° rotation
  - 90° rotation
- Use `cv2.getRotationMatrix2D()` and `cv2.warpAffine()`
- Display original and rotated images

Program

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Load the image
image = cv2.imread('spider.png')

# Display original image
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis('off')
plt.show()

# Image Translation
tx, ty = 100, 50
M_translation = np.float32([[1, 0, tx],
                            [0, 1, ty]])
translated_image = cv2.warpAffine(
    image, M_translation, (image.shape[1], image.shape[0])
)

plt.imshow(cv2.cvtColor(translated_image, cv2.COLOR_BGR2RGB))
plt.title("Translated Image")
plt.axis('off')
plt.show()

# Image Scaling
fx, fy = 5.0, 2.0
scaled_image = cv2.resize(
    image, None, fx=fx, fy=fy,
    interpolation=cv2.INTER_LINEAR
)

plt.imshow(cv2.cvtColor(scaled_image, cv2.COLOR_BGR2RGB))
plt.title("Scaled Image")
plt.axis('off')
plt.show()

# Image Shearing
```

```
shear_matrix = np.float32([[1, 0.5, 0],
                           [0.5, 1, 0]])

sheared_image = cv2.warpAffine(
    image, shear_matrix, (image.shape[1], image.shape[0])
)

plt.imshow(cv2.cvtColor(sheared_image, cv2.COLOR_BGR2RGB))
plt.title("Sheared Image")
plt.axis('off')
plt.show()

# Image Reflection
reflected_image = cv2.flip(image, 2)

plt.imshow(cv2.cvtColor(reflected_image, cv2.COLOR_BGR2RGB))
plt.title("Reflected Image")
plt.axis('off')
plt.show()

# Image Rotation
(height, width) = image.shape[:2]
angle = 45
center = (width // 2, height // 2)

M_rotation = cv2.getRotationMatrix2D(center, angle, 1)
rotated_image = cv2.warpAffine(
    image, M_rotation, (width, height)
)

plt.imshow(cv2.cvtColor(rotated_image, cv2.COLOR_BGR2RGB))
plt.title("Rotated Image")
plt.axis('off')
plt.show()

# Image Cropping
x, y, w, h = 100, 100, 200, 150
cropped_image = image[y:y+h, x:x+w]

plt.imshow(cv2.cvtColor(cropped_image, cv2.COLOR_BGR2RGB))
plt.title("Cropped Image")
plt.axis('off')
plt.show()
```

Developed By:

Name: S. Jana Shravin

Register No:

212224243003

Output

Cropped Image

Rotated Image

Reflected Image

Sheared Image

Scaled Image

Translated Image

Original Image

Image Translation

- Original image is displayed
- Translated image (shifted right and down) is displayed

Image Scaling

- Original image is displayed
- Downscaled image (0.5×) is displayed
- Upscaled image (2×) is displayed

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